Aerospace Mechanics & Physical Laws

The Balloon Rocket Adventure

To Every Action, An Opposite Reaction!

Newton's Third Law

Grade 3 Physics

30 Minutes

ISRO Anchor

The Srikarikota Countdown 🇮🇳

"Three... Two... One... Lift-off!"

👦 Vihaan asks Dada-ji:

Dada-ji look at the exhaust fire! It's pointing straight DOWN toward the ground...

...so why is the heavy rocket surging straight UP into the sky?

How does pushing DOWN help things go UP?"

Exhaust goes DOWN

Rocket goes UP??

Newton's Double Push Rule

Whenever an object pushes against something else, it receives a matching push right back — with the EXACT SAME STRENGTH, in the OPPOSITE DIRECTION.

👊 ACTION FORCE

The initial push applied — the element pushing backward or downward against an obstacle.

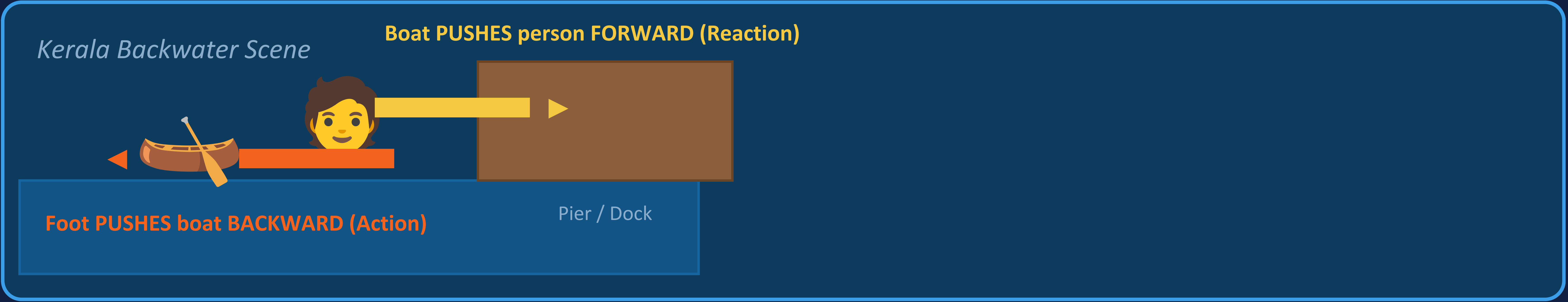
VS

🎯 REACTION FORCE

The structural response push — pushes back with equal strength in the OPPOSITE direction.

⚡ **IMPORTANT: Both forces happen at the EXACT SAME MOMENT — the reaction never waits!**

Everyday Example: Stepping Off a Naav 🛶



👣 ACTION

Your Foot Pushes

Your foot pushes BACKWARD hard against the wooden boat hull, propelling your body forward.

⛵ REACTION

Boat Pushes You

As you push the boat back into the water, the boat pushes YOUR BODY forward onto the dry pier.

Meet the Model: Straw-on-String Balloon Rocket 🎈

🔬 Lab Materials

〰️ Long String Path

Stretched tight between two walls — our guide track

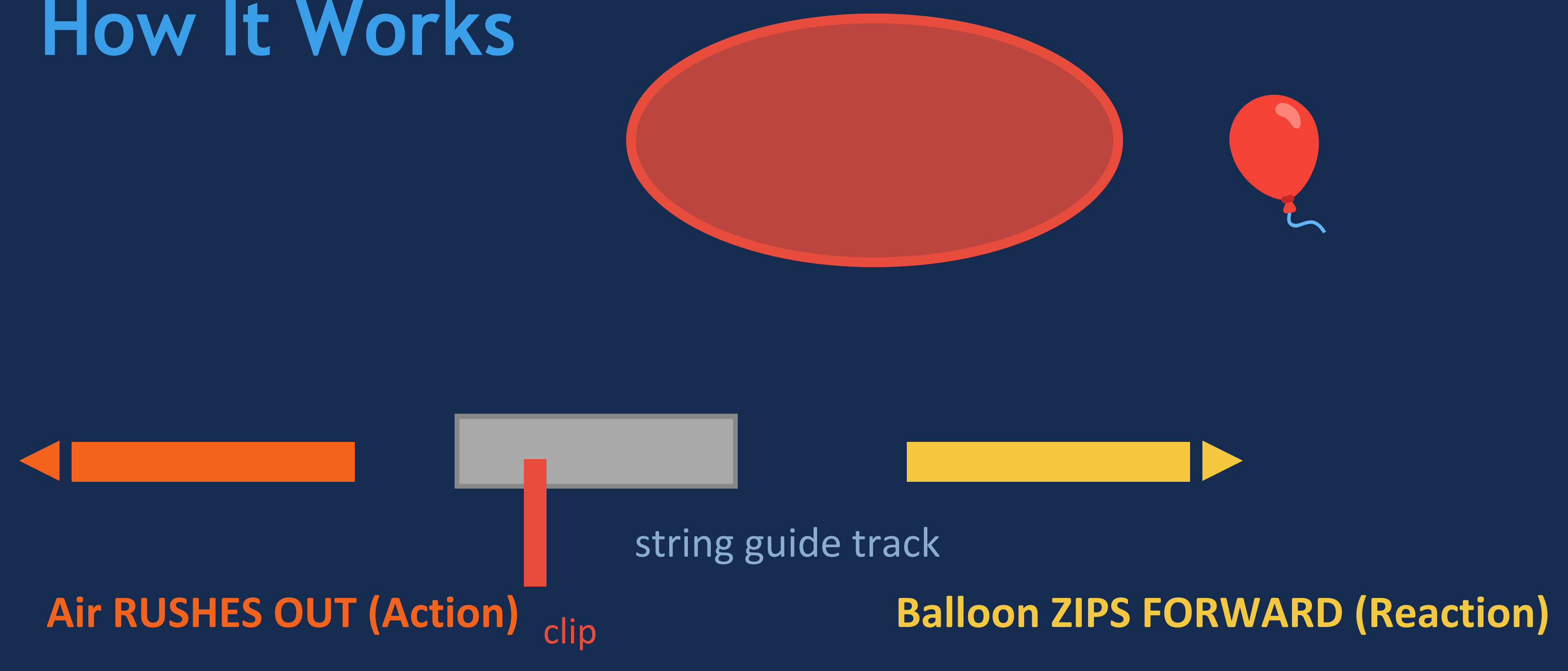
🥤 Plastic Drinking Straw

Threaded onto string — acts as the wheel bearing casing

🎈 Inflated Rubber Balloon

Taped to the straw — acts as fuel storage and engine casing

How It Works



Action Force (air backward)

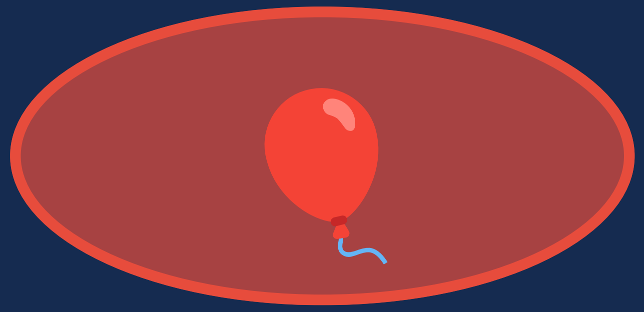
Reaction Force (balloon forward)

🖱️ **Click the scissors icon to release the clip and launch!**

Exploring the Model: Action & Reaction in Motion

PHASE 1 — ACTION

Exhaust Rushing Back

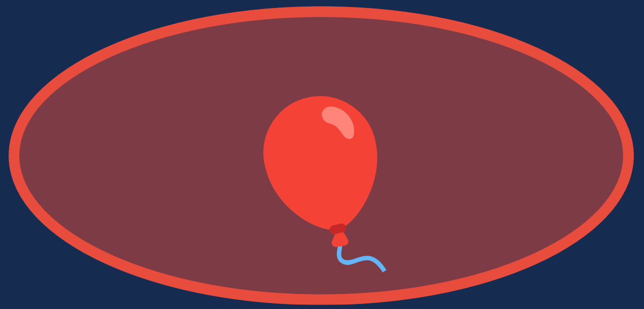


Air rushes BACKWARD

When the clip is removed, the stretched rubber walls squeeze inward, forcing the trapped air to escape through the open nozzle at HIGH SPEED.

PHASE 2 — REACTION

Traveling Forward



Balloon zips FORWARD

The escaping air pushes backward against the room, generating an IDENTICAL forward push against the balloon's inner wall — launching it along the string track!

Scaling Up: From Toy Balloons to ISRO Spaceships 🛸

🎈 Balloon Rocket —— Same Physics Rules —— 🚀 ISRO LVM3 Rocket

Balloon Rocket

Air pushes back (Action)

Balloon flies forward (Reaction)

ISRO LVM3 Rocket

Gas pushes DOWN (Action)

Rocket surges UP (Reaction)

✅ Vihaan's Answer: The rocket flies UP because its engines push gas DOWN — equal & opposite forces!

🎮 Guided Practice: ISRO Launch Director Game

Test Run	Nozzle Direction	Flight Outcome
Hangar Test 1	Angled Sideways ↗️	❌ Vehicle swerves off course, missing its target!
Hangar Test 2	Low Air Volume ↓	⚠️ Weak action push = very low, very short flight!
Hangar Test 3	Straight Down ⬇️	✅ **Max push = PERFECT vertical flight into space!**

🎯 ACTIVE MISSION: Turn the virtual nozzle dial STRAIGHT DOWN → Click GREEN LAUNCH button → Watch the rocket hit orbit!

🧪 Concept Review: Test Your Knowledge!

Question 1

If a rocket pushes exhaust gas toward the LEFT, which direction does the vehicle fly?

A) Backward (Leftward)

B) Forward (Rightward) ✓

C) Straight Downward

D) It Stays Still

Forces push OPPOSITE to the action direction!

Question 2

In the balloon rocket experiment, what serves as the primary ACTION force that sets the system in motion?

A) Air rushing out backward ✓

B) The balloon zipping forward

C) The long string guide track

D) The plastic straw housing

Air rushing BACKWARD = Action; Balloon flying forward = Reaction!

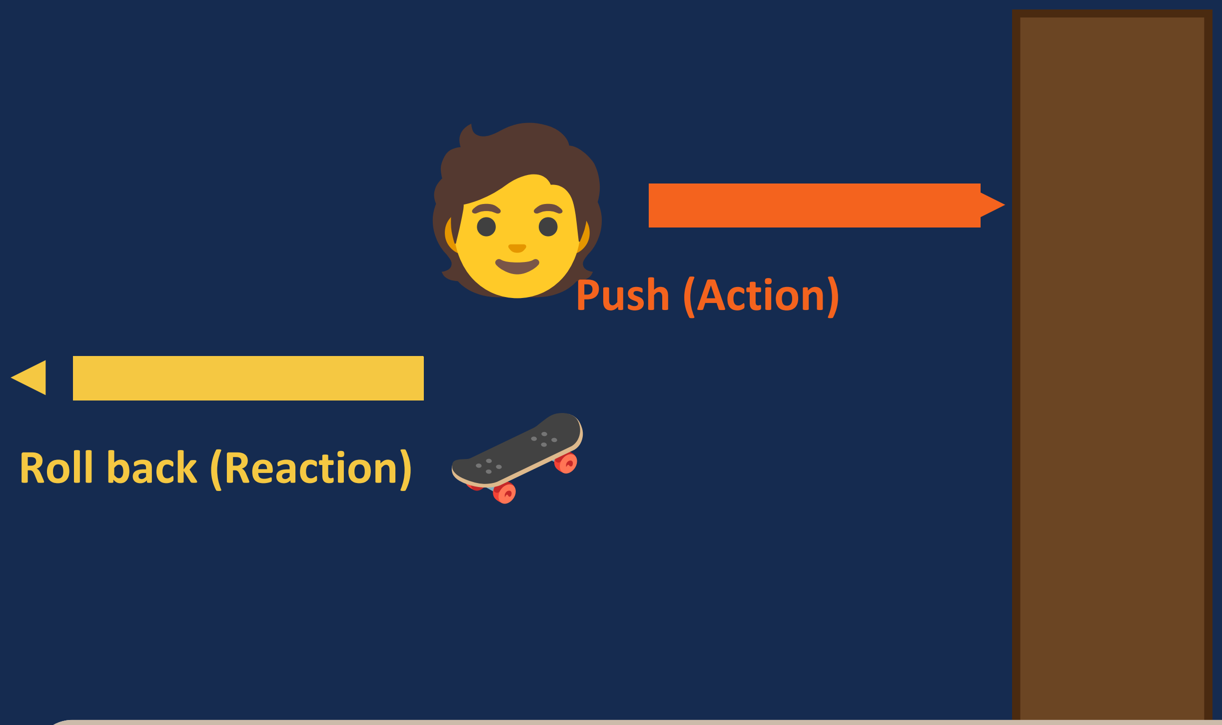
🛹 Home Physics Mission: The Skateboard Wall-Push Quest!

Your Mission Steps:

1. Go outside on a smooth playground with an adult helper.
2. Stand balanced on a skateboard, roller skates, or foot scooter.
3. Face a sturdy brick wall and place both palms flat against it.
4. Give the wall a firm, steady push forward away from your body.
5. Observe: your wheels roll BACKWARD away from the wall!

Always do this with an adult present for safety!

What You'll See:



📓 Sketch It!

= your push (Action)

= wheels roll (Reaction)

Mission Complete: The Double Push Takeaways

Forces Always Come in Pairs

In our world, you can never have a single push on its own. Forces always function as paired interactions.

Directions Are Perfectly Opposite

The reaction force always pushes back with equal strength in the EXACT OPPOSITE direction.

From Toys to Real Rockets

Whether launching a balloon or an ISRO LVM3 spacecraft, the same physics rules apply every time.

"By learning to map these opposite pushes, you've unlocked the core science secrets that help aerospace engineers guide heavy spaceships safely into outer space!"

Master Mission Controller Milestone Achieved!

Thank You!

Keep exploring the forces around you!